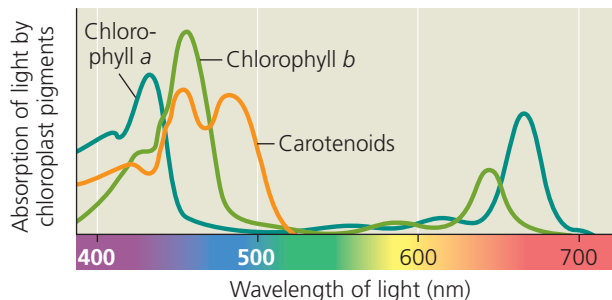


▼ Figure 10.9 Inquiry

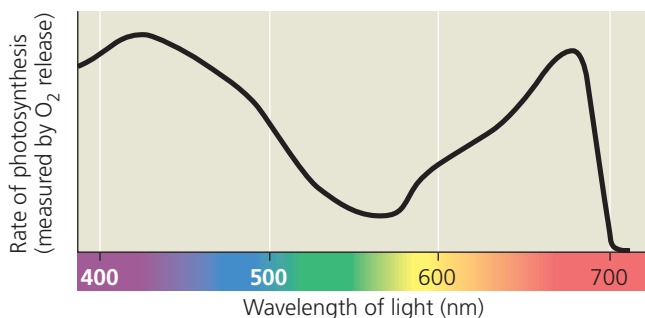
Which wavelengths of light are most effective in driving photosynthesis?

Experiment Absorption and action spectra, along with a classic experiment by Theodor W. Engelmann, reveal which wavelengths of light are photosynthetically important.

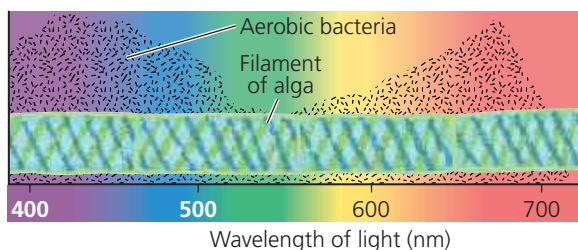
Results



(a) **Absorption spectra.** The three curves show the wavelengths of light best absorbed by three types of chloroplast pigments.



(b) **Action spectrum.** This graph plots the rate of photosynthesis versus wavelength. The resulting action spectrum resembles the absorption spectrum for chlorophyll *a* but does not match exactly (see part a). This is partly due to the absorption of light by accessory pigments such as chlorophyll *b* and carotenoids.



(c) **Engelmann's experiment.** In 1883, Theodor W. Engelmann shined light through a prism onto an alga, exposing different segments of the alga to different wavelengths. He used aerobic bacteria, which congregate near an oxygen source, to determine which segments of the alga were releasing the most O_2 and thus photosynthesizing most. Bacteria congregated in greatest numbers around the parts of the alga illuminated with violet-blue or red light.

Conclusion The action spectra, confirmed by Engelmann's experiment, show which portions of the spectrum are most effective in driving photosynthesis.

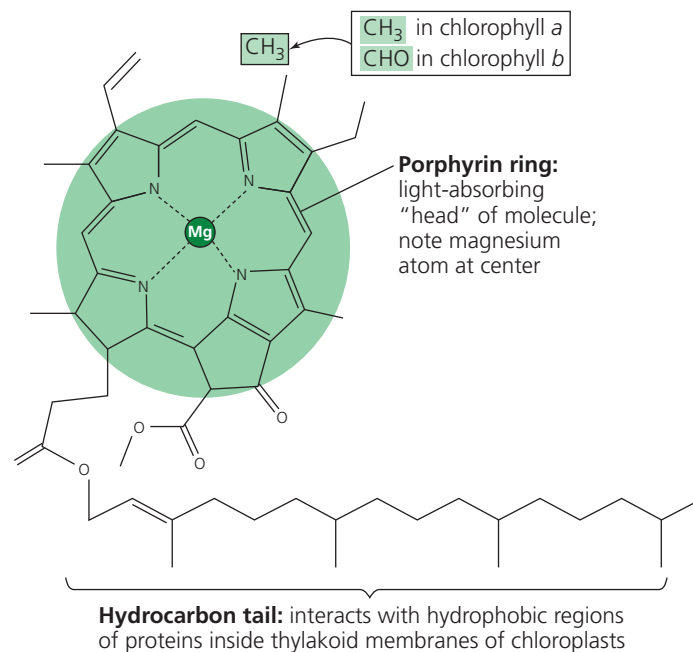
Data from T. W. Engelmann, *Bacterium photometricum*. Ein Beitrag zur vergleichenden Physiologie des Licht- und Farbensinnes, *Archiv. für Physiologie* 30:95–124 (1883).

➔ **Instructors:** A related Experimental Inquiry Tutorial can be assigned in Mastering Biology.

INTERPRET THE DATA According to the graph, which wavelengths of light drive the highest rates of photosynthesis?

▼ Figure 10.10 Structure of chlorophyll *a* and *b*.

Chlorophyll *a* and *b* differ in only one functional group:



➔ Mastering Biology

Animation: Space-Filling Model of Chlorophyll

Notice by comparing Figures 10.9a and 10.9b that the action spectrum for photosynthesis is much broader than the absorption spectrum of chlorophyll *a*. The absorption spectrum of chlorophyll *a* alone underestimates the effectiveness of certain wavelengths in driving photosynthesis. This is partly because accessory pigments with different absorption spectra—including chlorophyll *b* and carotenoids—broaden the spectrum of colors that can be used for photosynthesis. **Figure 10.10** shows the structure of chlorophyll *a* compared with that of chlorophyll *b*. A slight structural difference between them is enough to cause the two pigments to absorb at slightly different wavelengths in the red and blue parts of the spectrum (see Figure 10.9a). As a result, chlorophyll *a* appears blue green and chlorophyll *b* olive green under visible light.

In the last decade, researchers have identified two other forms of chlorophyll—chlorophyll *d* and chlorophyll *f*—that absorb higher wavelengths of light. In 2018, researchers grew a species of cyanobacterium called *Chroococcidiopsis thermalis*

► Colony of the cyanobacteria *Chroococcidiopsis thermalis*.

In this micrograph of cells becoming acclimated to higher wavelengths of light, cells that haven't yet acclimated and are still using chlorophyll *a* for photosynthesis appear purplish-pink, and cells that have acclimated and are using chlorophyll *f* appear yellow.

